L7b: Online SIM Estimation: Updating the Engine as Data Arrive

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Objectives for Today

Today we let the inputs move: L7a's engine recomputed its preferences daily, but its SIM intercepts and betas were frozen at the training values. We ask whether they were stable, replace the batch fit with an estimator that forgets at a stated rate, and compare frozen and updated engines on the realized year and on simulated futures.

Today's concepts

- Why frozen parameters can fail: rolling betas with bands, what they do and do not establish, and what a moving parameter does to the threshold, the basket, and the SIM covariance.
- The EWLS recursion: weighted loss, decay factor and half-life, three running moments, the solve, the residual scale, one prior.
- Frozen versus online, honestly: replay without look-ahead, a SIM scenario generator, the distributional scorecard, and what such an ensemble can and cannot show.

Lectures and Examples

Lecture:
CHEME-5660-L7b-Lecture-Online-SIM-Estimation-Fall-2026.ipynb

Example 1:
CHEME-5660-L7b-Example-EWLS-Replay-Fall-2026.ipynb

Example 2:
CHEME-5660-L7b-Example-Scenario-Ensemble-Fall-2026.ipynb

The first example estimates rolling betas with bands, runs the EWLS recursion, chooses and declares the half-life inside the training period, and replays the parameters through 2025; the second puts both parameter arms into the L7a engine on the realized year and on two thousand simulated futures.

Concept Review: The Batch SIM Fit and the Engine

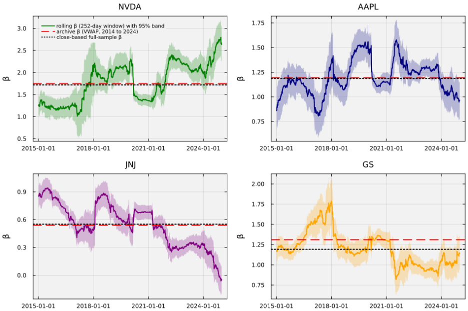
L6a estimated the SIM of asset i, $g_{i,t} = \alpha_i + \beta_i\, g_{M,t} + \varepsilon_{i,t}$ (g: annualized daily growth rates, ε: the residual, $t = 1, \ldots, N$ training days), by least squares: with response vector $\mathbf{y}$, design matrix $\hat{\mathbf{X}}$ (row t: $\mathbf{x}_t^\top = (1, g_{M,t})$), and $\theta_i = (\alpha_i, \beta_i)^\top$:

$$(\hat{\mathbf{X}}^\top \hat{\mathbf{X}})\hat{\theta}_i = \hat{\mathbf{X}}^\top \mathbf{y} \qquad \Longleftrightarrow \qquad \underbrace{\left(\sum_{t=1}^{N} \mathbf{x}_t \mathbf{x}_t^\top\right)}_{\text{Gram matrix}} \hat{\theta}_i = \underbrace{\sum_{t=1}^{N} \mathbf{x}_t\, g_{i,t}}_{\text{cross moments}}$$

Every day of 2014 counts as much as every day of 2024: one pooled slope.

- L7a engine: each day, from closes through $t-1$ (recent market growth $\tilde{g}_{M,t}$, market-state signal ξ_t), $\gamma_i(t) = \tanh(\alpha_i/\beta_i^{\xi_t} + \beta_i^{1-\xi_t}\tilde{g}_{M,t})$: sign chooses the basket ($\tilde{g}_{M,t} > -\alpha_i/\beta_i$), magnitude the share; trade at the close of t under $\mathcal{R}$, $\tau_{\max}$, $d_{\max}$, cost c.
- In L7a, (α_i, β_i) were the archive values on every 2025 day.

Why Frozen Parameters Can Fail: Rolling Bands



Refit the L6a regression on a one-year window ending on day t: $\hat{\beta}_i(t)$ with its classical pointwise band $\pm 1.96\,\mathrm{SE}$ (descriptive: iid assumptions, overlapping windows). It gives the motion a scale: a constant beta wanders inside such bands; an excursion of many band-widths lasting years would be unusual (NVDA: 1.0 to 2.8 against 1.75).

Why Frozen Parameters Can Fail: The Consequences

Rolling bands motivate testing an updating rule; they do not prove drift or promise improvement. A moving (α_i, β_i) acts on the engine in three places:

- The threshold $-\alpha_i/\beta_i$: the frozen thresholds sit within a few tenths per year of zero, so the basket hangs on whether $\tilde{g}_{M,t}$ is above or below a narrow band; an intercept that moves by a few tenths per year (its sampling error at any short memory) moves the threshold by more.
- The basket and the weights: the magnitude of $\gamma_i(t)$ moves through the lens $\beta_i^{\xi_t}$; a non-positive beta breaks the lens, so L7a's rule (assign to the non-preferred set) becomes necessary.
- The covariance matrix $\hat{\Sigma}_{g,\mathrm{SIM}} = s_{g,M}^2 \hat{\beta}\hat{\beta}^\top + \hat{\mathbf{D}}_g$ ($\hat{\beta}$ the betas, $\hat{\mathbf{D}}_g$ the diagonal of residual variances): with $s_{g,M}^2$ held fixed, the online betas and residual scales move it by 39% to 63% (Frobenius) after the identical first day.

None of this says updating helps: freezing is a choice with consequences.

EWLS: Weights, Loss, Decay Factor, and Half-Life

Exponentially weighted least squares keeps the L6a regression and changes the weight each day carries. Fix asset i; for days $s = 1, \ldots, t$ let $\mathbf{x}_s = (1, g_{M,s})^\top$, $y_s = g_{i,s}$ (inverse years), $\theta = (\alpha, \beta)^\top$. Choose a half-life $t_{1/2} > 0$ (trading days) and set the decay factor $\delta = 2^{-1/t_{1/2}} \in (0, 1)$, so $\delta^{t_{1/2}} = 1/2$. The estimate at time t minimizes:

$$\boxed{L_t(\theta) = \sum_{s=1}^{t} \underbrace{\delta^{t-s}}_{\text{weight of day } s} \left(y_s - \mathbf{x}_s^\top \theta\right)^2}$$

- An observation $t_{1/2}$ days old carries half today's weight; the total weight of the past is $\sum_{s \le t} \delta^{t-s} \to 1/(1-\delta) \approx t_{1/2}/\ln 2$, about 91 current-observation equivalents for a one-quarter half-life (a count of weight, not the variance-based effective sample size).
- For this data-only loss, ordinary least squares is the case $\delta = 1$.

EWLS: Sufficient Statistics and the Recursion

The loss is quadratic, so its minimizer solves a two-by-two system whose coefficients are weighted sums over the past, the weighted Gram matrix, cross-moment vector, and response second moment:

$$\mathbf{A}_t = \sum_{s\le t} \delta^{t-s}\mathbf{x}_s\mathbf{x}_s^\top \in \mathbb{R}^{2\times 2}, \qquad \mathbf{b}_t = \sum_{s\le t} \delta^{t-s}\mathbf{x}_s\, y_s \in \mathbb{R}^2, \qquad c_t = \sum_{s\le t} \delta^{t-s} y_s^2$$

$\nabla_\theta L_t = \mathbf{0}$ gives $\mathbf{A}_t\theta = \mathbf{b}_t$. Every weight of day s at time t is δ times its weight at $t-1$, and today enters with weight one, so each moment updates in one step:

$$\boxed{\mathbf{A}_t = \delta\,\mathbf{A}_{t-1} + \mathbf{x}_t\mathbf{x}_t^\top, \qquad \mathbf{b}_t = \delta\,\mathbf{b}_{t-1} + \mathbf{x}_t\, y_t, \qquad c_t = \delta\, c_{t-1} + y_t^2}$$

Solve $\mathbf{A}_t\hat{\theta}_{i,t} = \mathbf{b}_t$ directly (no inverse); the residual scale from the same moments, because the cross term cancels at the optimum:

$$\hat{\sigma}_{\varepsilon,i,t} = \sqrt{\frac{c_t - \hat{\theta}_{i,t}^\top \mathbf{b}_t}{[\mathbf{A}_t]_{11}}}, \qquad [\mathbf{A}_t]_{11} = \textstyle\sum_{s\le t} \delta^{t-s} \text{ (total weight; no dof correction)}$$

Three moments are sufficient: constant memory, constant work per day.

EWLS: The Prior, and the Units

$\mathbf{A}_0 = \mathbf{0}$ leaves the first days unidentified. Seed with the archive estimate $(\theta_{i,0}, \sigma_{\varepsilon,i,0})$ worth N_0 pseudo-observations through $\mathbf{M} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top] = \left(\begin{smallmatrix} 1 & g'_M \\ g'_M & s^2_{g,M}+g'^2_M \end{smallmatrix}\right)$ (training market moments):

$$\boxed{\mathbf{A}_0 = N_0\,\mathbf{M}, \qquad \mathbf{b}_0 = \mathbf{A}_0\,\theta_{i,0}, \qquad c_0 = \theta_{i,0}^\top \mathbf{b}_0 + N_0\,\sigma^2_{\varepsilon,i,0}}$$

- Returns the prior exactly before any update; after t updates its weight is $\delta^t N_0$: it decays like the data. The seeded estimate minimizes $L_t(\theta) + \delta^t N_0(\theta - \theta_{i,0})^\top \mathbf{M}(\theta - \theta_{i,0})$ (advanced notebook); ridge (L6a advanced) is a different modification.
- Units: annualized daily growth rates from closes; α, $\hat{\sigma}_\varepsilon$ in inverse years, β dimensionless, δ per observation, $t_{1/2}$ in trading days, N_0 in pseudo-observations. $\mathbf{M}$ is VWAP-based (4.6 vs 7.5 on closes): 63 pseudo-observations carry the slope information of about 39 close-based days.

Algorithm: EWLS SIM Update (Online)

Initialize: the prior for each asset, the archive market moments, N_0, $t_{1/2}$ (hence δ); seed $(\mathbf{A}_0, \mathbf{b}_0, c_0)$, the same $\mathbf{A}_0$ for every asset.

For each new trading day t:

1. After the close, observe $g_{M,t}$ and $g_{i,t}$ for every asset; form $\mathbf{x}_t = (1, g_{M,t})^\top$.
2. Update every asset's moments: $\mathbf{A}_t \leftarrow \delta\mathbf{A}_{t-1} + \mathbf{x}_t\mathbf{x}_t^\top$, $\mathbf{b}_t \leftarrow \delta\mathbf{b}_{t-1} + \mathbf{x}_t g_{i,t}$, $c_t \leftarrow \delta c_{t-1} + g_{i,t}^2$.
3. Solve $\mathbf{A}_t\hat{\theta}_{i,t} = \mathbf{b}_t$ and compute $\hat{\sigma}_{\varepsilon,i,t}$; if numerically unidentified, keep the previous estimate.
4. Hand $(\hat{\alpha}_{i,t}, \hat{\beta}_{i,t}, \hat{\sigma}_{\varepsilon,i,t})$ to the engine for use on day $t+1$, never on day t.

Output: the parameter paths and the final moments. Package: `ewls_init`, `ewls_update!`, `ewls_path`.

Example 1: CHEME-5660-L7b-Example-EWLS-Replay-Fall-2026.ipynb

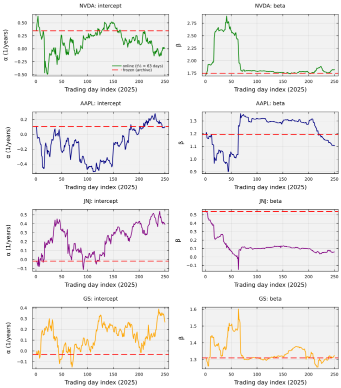
Chronological Replay: Declaring the Half-Life

The half-life is the one knob, chosen inside the training period by a walk-forward comparison: for each $t_{1/2} \in \{21, 63, 126, 252, \infty\}$, run the recursion over 2014 to 2024 seeded on the first 252 growth rates alone, and on every day t of 2020 to 2024 predict from the parameters through $t-1$:

$$\hat{g}_{i,t\,|\,t-1} = \hat{\alpha}_{i,t-1} + \hat{\beta}_{i,t-1}\, g_{M,t} \qquad \text{(conditional on } g_{M,t}\text{: a test, not a forecast)}$$

- One quarter wins on aggregate (1.9% below no forgetting) and for seven of thirteen firms. Declared: $t^{\star}_{1/2} = 63$ trading days; nothing re-tunes it. The criterion scores $\hat{\alpha} + \hat{\beta} g_M$, not the threshold classification the engine acts on.
- The replay: frozen reads the archive every day; online restarts on the first 2025 day from the archive prior ($N_0 = 63$), updates on close-based growth rates, and on day t reads the parameters that stood before day t's observation. Parameters diverge from day 2; trades from day 22.

Chronological Replay: What Changed in 2025



- Online betas moved by half a unit or more, intercepts by several tenths per year (the least identified parameter at a one-quarter memory); the arms disagreed on at least one firm on 101 of 250 days; empty basket 12 days (online) against 57; JNJ's beta non-positive for three April days (the rule applied).
- Updating hurt: online monthly 1.33 against frozen 1.53 (daily 1.26 against 1.40), twice the turnover, one breaker firing.
- One draw, not a probability. Where the gap opened: next frame.

Example 2: CHEME-5660-L7b-Example-Scenario-Ensemble-Fall-2026.ipynb

Chronological Replay: Two Episodes, One Rerun

Example 2 pins the gap to dates, holdings, and one rerun:

- Spring: the online intercept of JNJ put that firm alone into a basket the frozen thresholds left empty (98% of the online book on March 31, ahead 3.5%); the April fall left the online arm 8.9% down and about 58 dollars behind, a gap near 40 to 56 dollars until November.
- November: both arms peaked on November 12 and drew down to November 20, frozen by 9.0%, online by 10.1%, a tenth over the limit: the breaker liquidated the online book, its lock covered the December rebalance day, and the book sat in cash to year end.
- Counterfactual: breaker disabled, the same online engine ends at 1.50; the firing is 166 of the 202 dollars of the gap.
- Not “the rule, not the estimator”: the estimates changed the holdings, the holdings crossed the threshold, the discontinuous rule amplified a tenth of a percent into a quarter of a year in cash; one path cannot separate them.

The Scenario Ensemble: The SIM Generator

To compare policies across many possible years, draw futures from the model the engine carries: from the decision-date closes, for $t = 1, \ldots, N$, independently across days and assets, in inverse years,

$$\boxed{g_{M,t} \sim \mathcal{N}(g'_M,\, s^2_{g,M}), \qquad \varepsilon_{i,t} \sim \mathcal{N}(0,\, s^2_{g,\varepsilon,i}), \qquad g_{i,t} = \hat{\alpha}_i + \hat{\beta}_i\, g_{M,t} + \varepsilon_{i,t}}$$

then compound with $\Delta t = 1/252$ yr: $S_M(t) = S_M(t-1)\, e^{g_{M,t}\Delta t}$, $S_i(t) = S_i(t-1)\, e^{g_{i,t}\Delta t}$; the shared draw makes the assets co-move as $\hat{\Sigma}_{g,\mathrm{SIM}}$ says.

- Units: draw daily growth rates in annualized units; Δt enters once, in the exponent: a one-day log return with mean $g'_M \Delta t$, variance $s^2_{g,M} \Delta t^2$ (L4b, L5a).
- Scenario engine, not validation: Gaussian, independent residuals, constant parameters, VWAP-smoothed moments; same generator (L5b): the frozen parameters are the generator's own, so online estimates track noise; the benefit under drift is out of reach.

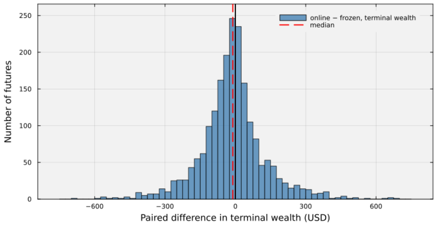
The Scenario Ensemble: The Distributional Scorecard

With n futures of N days ($T = N\Delta t$; $W_{\mathcal{P}}$ the account's wealth, g_f the risk-free rate), terminal wealths sorted $W_{(1)} \leq \cdots \leq W_{(n)}$, drawdowns $D^{(p)}$, tail level 5%, $k = \lceil 0.05\, n \rceil$:

- Median terminal wealth, median NPV: the typical outcome; did it beat cash at g_f.
- Fail rate: fraction of futures with $W_{\mathcal{P}}^{(p)}(T) < W_{\mathcal{P}}(0)e^{g_f T}$ (negative NPV).
- Lower quantile at 5%: the order statistic $W_{(k)}$, so at least 5% of futures end at or below it (with $n = 2000$: the hundredth smallest, a definite number).
- Tail mean at 5%: $\frac{1}{k}\sum_{j \leq k} W_{(j)}$, the worst 5% on average.
- Drawdown median and 5% upper cutoff: the median and the k-th largest $D^{(p)}$.

Paired: both arms run on the same futures, so compare online minus frozen terminal wealth per future (median, spread, fraction positive); pairing holds the market and residual draws fixed and leaves the policy contrast.

The Scenario Ensemble: Two Thousand Futures



- Frozen, monthly: median 1.14 times initial wealth, fail rate 27%, 5% quantile 0.93, tail mean 0.89, median drawdown 10.6%. The realized 2025 (1.53) was a favorable draw.
- Online, monthly: a little worse on every column (median 1.12, fail 31%, quantile 0.90); SPY is close to both.
- Paired: median about -11 on a thousand, standard deviation about 140; online ahead on 44% of futures, below half: the observed cost of updating with no drift to learn.

Example 2: CHEME-5660-L7b-Example-Scenario-Ensemble-Fall-2026.ipynb

Validation Gates: A Pointer

The comparisons of today, frozen against online on a realized path and on an ensemble, become in L15a a set of deployment gates: pass-or-fail rules with stated limits (a median terminal wealth that must beat the frozen baseline, a fail rate and a tail that must not exceed set values, a drawdown and a turnover budget), evaluated on the same paired ensemble and reported before an online engine is allowed to replace its frozen ancestor.

Nothing today is such a gate; L15a turns these comparisons into pass or fail.

Optional Advanced Material

Advanced: advanced/online_learning/
CHEME-5660-L7b-Advanced-EWLS-Recursion-Fall-2026.ipynb

Optional, not a prerequisite for L8b. Derives the weighted normal equations and the three sufficient statistics from the exponentially weighted loss, shows the one-step recursion, solves for the parameters and the residual scale (with the cancellation that makes the residual formula work), and proves that the prior seeding returns the prior exactly and that the seeded recursion minimizes the data loss plus a decaying prior-centered quadratic.

Notebook: lectures/week-7/L7b/advanced, also linked from the lecture notebook.

Summary

Today: pooled parameters that mask time variation; the EWLS recursion; a replay under a declared half-life; a SIM scenario ensemble and its scorecard.

Key takeaways

- A frozen parameter is a modeling choice, and rolling bands give the motion a scale: a moving intercept or beta moves the threshold, the basket, and the SIM covariance.
- EWLS is least squares with a memory, and everything about it is stated: decay and half-life, sufficient moments, a two-by-two solve, a prior that decays like the data, a half-life chosen inside the training period.
- Compare policies on many futures, and say what the futures are: on 2025 the online engine finished behind, mostly through one breaker firing; on two thousand futures the frozen engine's typical outcome was far below its 2025 result and online paid a small price with no drift to learn; not validation.

Next, L8b: option contracts. Frozen versus online returns in L15a as gates.

Today: parameters that move, an estimator that forgets at a stated rate, and a scorecard read across many futures.